Algorhythmics and Media Concepts: Ada Lovelace's Notes to "Sketch of the Analytical Engine"

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One of the more often-cited passages from Ada Lovelace's notes to Luigi Menabrea's "Sketch of the Analytical Engine" (1843) is her meditation on the computational possibilities of electronic music: "Supposing, for instance, that the fundamental relations of pitched sounds in the science of harmony and of musical composition were susceptible of such expression and adaptations, the engine might compose elaborate and scientific pieces of music of any degree of complexity or extent" (Notes 694). Lovelace's description of electronic music embraces digital simulation, in which analyzed musical signals are mathematically recreated to produce new and unique compositions. Lovelace's speculation both underscores her interest in music and distinguishes her understanding of the potential of the Analytical Engine

Abstract: Ada Lovelace's notes to Luigi Menabrea's "Sketch of the Analytical Engine" (1843) constitutes a significant development in the history of what John Guillory calls "the media concept." Standing in contrast to her mentor Charles Babbage, who saw mediation as that which is in between input and output, Lovelace distinguishes the material operation of the machine from the digital simulation performed by its manipulation of numbers. Her image of computing is a musical composition, in which mathematical processes are seen as various materials and temporalities orchestrated together. I link Lovelace's work to the "algorhythmics" of Shintaro Miyazaki and the "time-criticality" of Wolfgang Ernst. I also perform a grammatological analysis of her program for computing Bernoulli numbers as a demonstration of how she sees the distinct but orchestrated domains of material and number working together. The Bernoulli program, along with her reflections on the media implications of the Analytical Engine, enabled Lovelace to develop her own media concept decades before the publication of Walter Benjamin's "The Work of Art in the Age of Mechanical Reproduction" (1935).

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from that of its designer, Charles Babbage. Lovelace was well-versed in musical composition from a young age. Early records of her schedule by her mother, Lady Annabella Byron, include daily musical instruction.[1] During her correspondence with Mary Somerville in 1837, she wrote in a letter dated June 22, "I play four or five hours generally, & never less than three" (emphasis original). Furthermore, Lovelace was known as a great fan of opera, particularly Vincenzo Bellini's *Norma* (1831).[2] These experiences, along with the references to electronic music in her notes, point to a musical perspective on computer programming. I present Lovelace's perspective in her notes on Luigi Menabrea's "Sketch of the Analytical Engine" with reference to media archaeological theories of time and rhythm to show how the notes theorize what John Guillory would later call the "media concept" (*passim*). This media concept is, in Guillory's words, "a response to the proliferation of new technical media—such as the telegraph and the phonograph—that could not be assimilated to the older system of the arts" (321). Lovelace's work maps computing processes from the movement of material components to their numerical and algorithmic abstraction, and to the properly digital simulations composed by such abstractions. I argue that Lovelace moved away from Babbage's understanding of mediation, which James Brooke-Smith identifies as a general sense of "in-betweeness," to a technical conception of a channel mediating a signal. Lovelace's application of the media concept to the Analytical Engine enabled her to speculate about electronic music decades before it was technologically realized.

Brooke-Smith points out that "in-betweenness" linked Babbage to Romantic poets like William Wordsworth, whom Andrew Burkett sees as engaging with a sense of mediation that was more "capacious, labile, open-ended" than those emerging later in the century (139). I contextualize these respective arguments within the nineteenth-century discourse of a shifting media concept, which Guillory argues emphasized physics and engineering over aesthetics. Richard Menke references this discourse when he mentions the "vernacular media theory" (21) of Victorian authors, whose metadiscursive implications Paul Fyfe sees as deeply embedded in the development of the digital humanities as a field.[3] Lovelace's speculations add to Babbage's concept by referencing the materialities and processes of the machine as it simulates electronic music.

Lovelace's work contextualizes digital media according to the time-critical operations of what media archaeologists call "algorhythmics." Shintaro Miyazaki's work on algorthymics sees digital media as an "interplay, orchestration, and synthesis" of algorithms, signals, and materialities ("Algorhythmics"). Algorithms manipulate numbers; these numbers reference a technical processing of signals; these signals move through a rhythmic, timed, and orchestrated

material operation that Wolfgang Ernst calls "time-criticality" (Young 10).[4] Ernst discusses Fourier Wave Analysis as an example of time-criticality, in which a sound wave is divided into discrete sines and cosines for the purpose of mathematical analysis and electronic sound processing. He quotes Gottfried Wilhelm Leibniz's reflection about the intricate division of sound by the ear: "in hearing the roar of the sea, I hear—though without distinguishing them—the individual little noises of the waves out of which the total noise is made up" (qtd. in *Sonic Time Machines* 50). As media archaeological scholars point out, electronic sound works in a similar manner by recording, replaying, and digitally rearranging soundwaves as part of media processing. Miyazaki articulates this process in terms of a "spatiotemporal actualization, unfolding, and articulating" that digitally transforms signals into new compositions (*Counter-Dancing Digitality* 9). For Lovelace the "fundamental relations" of sounds are numerical in nature, and their rhythmic manipulation by the materiality of the Analytical Engine constitutes a time-critical process which makes possible the development of electronic music and general computing in the twentieth century.

I suggest that Lovelace's expertise in both music and mathematics helped her develop a technical media theory of computer operation in her notes to Menabrea's work on Babbage. As Imogen Forbes-Macphail puts it, for Lovelace, "mathematical symbols and operations are manifested in 'matter and movement' simultaneously, reciprocally, the material Engine becomes capable of signifying" (153). As we'll see in the next section, Babbage's sense of in-betweenness focused on the Engine's necessary inputs and outputs for rapidly delivering a commodity. This essay depicts how Lovelace's notes move from the media-less theories of Babbage to an early articulation of Guillory's media concept by tracing them from Babbage's theories when designing the Engine to Lovelace's notes. Lovelace's descriptions of rhythmic composition and numerical abstraction distill Babbage's capacious sense of media to a more technical and situated process that enabled Lovelace to theorize digital simulation and general computing.

Babbage's Engines

Babbage's *On the Economy of Machinery and Manufactures* (1832) elucidates the central aim of his Engines as providing the maximum return for the investments of British shipping industrialists. The Engines provided Babbage a logic in which labor could be exchanged and automated. Babbage saw labor, in Seb Franklin's words, as a map of "human potential onto machine actions" (24). Babbage's machines were the ultimate technological expression of the capitalist division of labor, in which labor was modularized with increasing granularity.

Babbage spends a chapter in *On the Economy* using the Difference Engine as an "illustration of his theories" (13). Yet despite his talk about efficiency and value, he failed to complete a full-sized prototype of the Difference Engine before the British government pulled his funding. As I have detailed elsewhere, in my article on Babbage and the Difference Engine, the replica originally produced for Babbage's 200th birthday in 1991 by the Computer Science Museum took over seventeen years and many mishaps to create—costing well over one million dollars.[5]

Babbage also never completed a prototype of his designs for the Analytical Engine. Looking briefly at the major components of the machine demonstrates how much it differs from the calculating functions of the Difference Engine. A prototype of the Difference Engine No. 2 on display at the Computer History Museum in Mountain View, California measures about eleven feet long, weighs about five tons, and is composed of over eight thousand parts.[6] By contrast, Allan Bromley says that the "Analytical Engine would have stood about 15 feet high. The mill would have been about 6 feet in diameter, and the store would have run 10–20 feet to one side. The Analytical Engine would therefore have been about the size and weight of a small railway locomotive" (198). Crucially, the Analytical Engine could store numbers, loop values until a particular condition was met, or choose between functions based upon the solution of a prior function. It had a hard drive, called "The Store," which was made up of several round columns on the right part of the design, and mechanically held number values on either side of what was labelled "The Rack." The "Counting Apparatus" was located to the left of the upper set of "Store Axes" and included a "Variable Card Reader" that would allow the user to include different values in punchcards for a given equation. This card reader sent the value to the "Ingress Axis," which prepared it for the Mill. "The Mill" worked like a central processing unit made up of a giant gear train in the center of the device. Various gear trains and columns made up the outward boundary of the Mill, including the "Boarding Carriage," "Tail Axis," "Anticipating Carriage," "Head Axis," "Reducing Apparatus," "Central Wheels," and "Table Axis." These were accompanied by an "Operation Card Reader," which determined whether an operation added, subtracted, divided, or multiplied; an "Egress Axis," which took the iteration produced by the Mill and printed it on tables; and a number of barrels, which ran completing, branching, or looping operations.[7]

Christopher Hollings, Ursula Martin, and Adrian Rice note that Babbage approximated it would take the Engine about "three minutes to multiply two 20-digit numbers" (76). However dazzling, Babbage imagined the Analytical Engine as primarily a calculator that would save enormous amounts of time and capital. He designed punchcards for the Analytical Engine from the cards

developed for Joseph-Marie Jacquard's automated loom: a programmable weaving system developed in 1804. Jacquard's cards copied textile patterns developed by artisans and automated them to make their manufacturing more efficient. Babbage adopted Jacquard's punchcards to perform two separate functions: to "direct the nature of the operation to be performed [and to] direct the particular variables" used in the Engine's calculations (*Passages* 120). While the variables and operations varied, Babbage still contextualized these elements within a sense of mediation that was numerical: he leveraged fixed mathematics to articulate the calculating process operating between the machine's input and output, and he understood algorithms as primarily devoted to the production of a numeric result. Lovelace clearly found the breadth of Babbage's mechanical vision compelling, yet she added abstraction and simulation to construct a more technical sense of mediation and envision many more possibilities than numbers and commodities.

Lovelace's Algorhythms

In an October 29, 1851, letter to her mother, Lovelace imagines numbers "marching in irresistible power to the sound of Music. . . . But then, what are these numbers? There is a riddle." The substance of these marching numbers might be a mystery, but this mystery has little impact upon the operations of the machine. By privileging number over substance, Lovelace expresses a technical sense of mediation that Guillory defines as focusing on "a process rather than an object" (342). Technical mediation also enabled Lovelace to escalate her argument to suggest that the Engine "might act upon other things besides number" and that it could weave "algebraical patterns just as the Jacquard loom weaves flowers and leaves" (Notes 697). When Lovelace describes the process of Babbage's machine in Note D of "Sketch of the Analytical Engine," she writes that the process includes "several distinct *sets of effects* going on simultaneously; all in a manner independent of each other, and yet to a greater or lesser degree exercising a mutual influence" (710, emphasis original). Her point is primarily statistical: that the Engine can perform more complex mathematical calculations that would ordinarily be impossible to compute at scale. Yet, like her militaristic fantasy of marching numbers, the emphasis on the Engine's material operation emphasizes attention to process over object. Distinct instruments or parts of the Analytical Engine perform independently, yet they also mutually implicate one another into the harmonies, rhythms, and other musical forms of computer operation.

Menabrea's "Sketch of the Analytical Engine" promoted Babbage's designs when the latter sought funding alternatives to the British government.

As Menabrea puts it in the first section: "the illustrious inventor having been kind enough to communicate to me some of his views on this subject during a visit he made at Turin [in 1840], I have, with his approbation, thrown together the impressions they have left on my mind" (671). Babbage recounts suggesting that Lovelace translate and add notes to Menabrea's memoir, explaining that they "discussed together the various illustrations that might be introduced: I suggested several, but the selection was entirely her own" (*Passages* 136). Hollings, Martin, and Rice mention that of the sixty-six pages devoted to Menabrea's article, "forty-one are [Ada Lovelace's] appendices" (77). Despite Babbage's own testimony, the division of labor between them remains controversial, with many prominent Babbage scholars dismissing Lovelace's contributions. Thomas Misa shows that Lovelace transformed Babbage's mathematics "into a precise series of elemental additions, multiplications, and substitutions," suggesting that she implemented them into a computing process that could be programmed into the Analytical Engine (14). For me, Lovelace's implementation recalls the division of materials and processes that is crucial to her account of mediation.

Lovelace understands the Analytical Engine as, in Miyazaki's words, "a machine that makes time itself logically controllable and, while operating, produces measurable time effects and rhythms" ("Algorhythmics"). I identify these material processes as "rhythmic" to reference not only the way time effects are measured through physical gears and their movement as the Engine computes algorithmic processes but also Lovelace's musicality. Lovelace understands an operation as "any process which alters the mutual relation of two or more things," while imagining "combinations unlimited in variety and extent if the deductions from the primary relations be carried on far enough" (Notes 693). These patterns could, themselves, impact human thinking as they throw into relief new realities that were invisible before the use of the machine: "we might even invent laws for series or formulæ in an arbitrary manner, and set the engine to work upon them, and thus deduce numerical results which we might not otherwise have thought of obtaining" (722). In the context of her reflections on electronic music, understanding any signal is akin to understanding their numerical relationship with other signals. If one understands that relationship, one can express that relationship mathematically. And if one can program the relationship through the material operations of the Analytical Engine, then the relationship becomes the object of process and further operation. Simulated music emerges once we understand that relationship and reproduce it. Lovelace applies her insight about musical tonality to every elemental material process. She describes general computing as the abstraction of

these processes, which require the breaking down of phenomena into discrete components whose number can be acted upon by the Engine.

The program that Lovelace published generating Bernoulli numbers is a powerful example of a mathematical relationship embodied by a material operation whose abstraction provides another set of relations that could be acted upon by the Engine. Misa explains that Bernoulli numbers aided in calculating "the summation of powers ($1^n + 2^n + 3^n + 4^n + \cdots$). With their assistance Jacob Bernoulli computed an immense number 91, 409, 924, 241, 424, 243, 424, 241, 924, 500, in (as he claimed) 'less than half of a quarter of an hour'" (21). Johannes Faulhaber's formula calculates the polynomial sequence for the summation of powers, significantly reducing the amount of work required.[8] Lovelace's Bernoulli program could conceivably compute the entire sequence to infinity, and yet her work was limited by the physical constraints of the Analytical Engine. As such the program she appended to "Sketch of the Analytical Engine" only calculates up to the eighth Bernoulli number, or $1^n + 2^n + \cdots 8^n$ (fig. 1).

The table is a spatial representation of a set of temporal processes for the purpose of illustrating how the Analytical Engine operates in a time-critical manner. The values represent different moments of the ongoing algorithmic nature of Lovelace's program. Let's begin calculating one of the Bernoulli

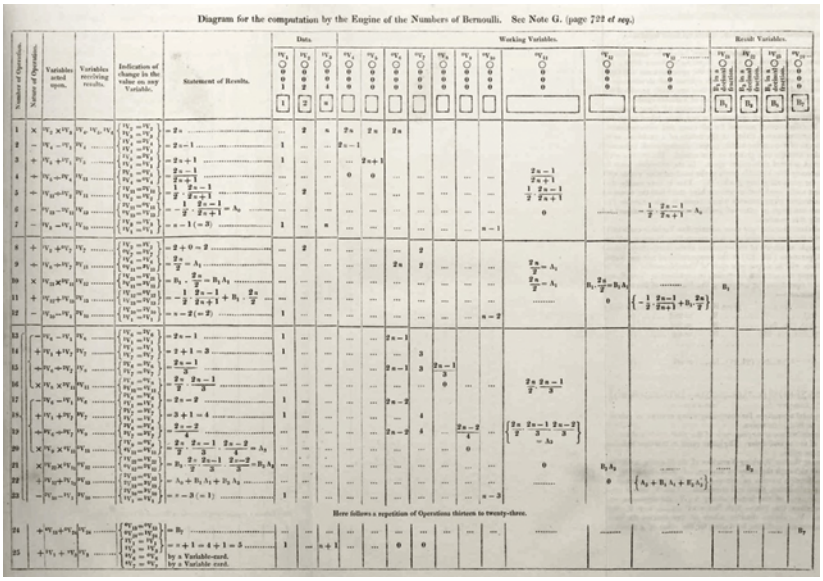

Fig. 1. Ada Lovelace's Bernoulli Program. From Note G, Notes by the Translator, "Sketch of the Analytical Engine Invented by Charles Babbage Esq." by Luigi Menabrea with notes by Ada Lovelace, 1843.

numbers to demonstrate how the process works. Greg Alt offers a concise summary of Lovelace's program by simplifying the table to exclude some of the more confusing and redundant columns, and my calculation uses his summary as a guide. He brackets the "working variables" while separating the first half of the table from the second half. The result looks something like this (fig. 2a–b).

Alt explains that "basic constants are set with $v1 = 1$ and $v2 = 2$, and $v3 = n$ is the input to the calculation, expected to be an arbitrary positive integer." Operation 1 would give us $v2 \ x \ v3 = 2n$ or $2 \times n = 2n$, and $2n$ is stored in $v4$, $v5$, and $v6$ respectively. $2n$ is listed under the "Statement of Results" of Operation 1 of Alt's table. Note that Operation 1 ends there and does not repeat. Operation 2 decrements $v4$ by one (in the table $v4 - v1 = 2n - 1$), and Operation 3 increments $v5$ (or $v5 + v1 = 2n + 1$). You can also see both formulas in the results column of Alt's table. I list the next few of Lovelace's subsequent operations below. Operation 4 asks us to divide $v4$ by $v5$, or:

$$\frac{v4}{v5} = \frac{2n-1}{2n+1}$$

The results of Operation 4 are stored in $v11$, which is immediately made part of Operation 5. Operation 5 divides $v11$ by $v2$:

$$\frac{v11}{v2} = \frac{1}{2} \times \frac{2n-1}{2n+1}$$

This is just the results for the first five operations using very simple starting set values. Alt notes that the rest of operations 6 through 25 are also basic math. The program becomes more complicated as we start to understand that the Bernoulli program also incorporates looping and conditional branching. The looping operation takes the value of $v10$ from previous operations, decrements, and then loops back to Operation 1.[9] If $v10 = 0$ at the end of the operations, the looping stops.

The Analytical Engine also supports conditional branching, creating different flows of operations depending upon certain conditions. You can see Alt's visual summary of these flows:

(1...7), (24, 25).....*gives* $B1 = 1st$ *number*; (*n being* = 1).

(1...7), (8...12), (24, 25)..... $B3 = 2nd$ *number*; (*n being* = 2).

(1...7), (8...12), (13...23), (24, 25)..... $B5 = 3rd$ *number*; (*n being* = 3).

(1...7), (8...12), 2(13...23), (24, 25)..... $B7 = 4th$ *number*; (*n being* = 4).

.....

.....

(1...7), (8...12), $\Sigma(+1)\,n - 2$(13...23), (24, 25)... $B2n - 1 =$ *nth number*; (*n being* = *n*).

Data.				Result Variables.			
v1	v2	v3	...	v21	v22	v23	v24...
0	0	0	...	B1 in a	B3 in a	B5 in a	0
0	0	0	...	decimal	decimal	decimal	0
1	2	4	...	fraction	fraction	fraction	0
[1]	[2]	[n]	...	[B1]	[B3]	[B5]	[B7]

Number of Operation.	Variables acted upon.	Variables receiving results.	Statement of Results.
1	v2 * v3	v4,v5,v6	= 2n
2	v4 - v1	v4	= 2n-1
3	v5 + v1	v5	= 2n+1
4	v5 / v4	v11	= (2n-1)/(2n+1)
5	v11 / v2	v11	= 1/2*(2n-1)/(2n+1)
6	v13 - v11	v13	= -1/2*(2n-1)/(2n+1) = A0
7	v3 - v1	v10	= n-1 (=3)
8	v2 + v7	v7	= 2+0 = 2
9	v6 / v7	v11	= 2n/2 = A1
10	v21 * v11	v12	= B1*2n/2 = B1A1
11	v12 + v13	v13	= -1/2*(2n-1)/(2n+1)+B1*2n/2
12	v10 - v1	v10	= n-2 (=2)
13	v6 - v1	v6	= 2n-1
14	v1 + v7	v7	= 2+1 = 3
15	v6 / v7	v8	= (2n-1)/3
16	v8 * v11	v11	= 2n/2*(2n-1)/3
17	v6 - v1	v6	= 2n-2
18	v1 + v7	v7	= 3+1 = 4
19	v6 / v7	v9	= (2n-2)/4
20	v9 * v11	v11	= 2n/2*(2n-1)/3*(2n-2)/4 = A3
21	v22 * v11	v12	= B3*2n/2*(2n-1)/3*(2n-2)/3 = B3A3
22	v12 + v13	v13	= A0 + B1A1 + B3A3
23	v10 - v1	v10	= n-3 (=1)
Here follows a repetition of Operations thirteen to twenty-three.			
24	v13 + v24	v24	= B7
25	v1 + v3	v3	= n+1 = 4+1 = 5; by a Variable-card. by a Variable card.

Fig. 2a–b. Greg Alt's simplification of Ada Lovelace's program. From "Lovelace's Program, Part 1: First Look at the First Program," *Pair Debugging with Ada Lovelace*, 18 Aug. 2024, https://pairdebuggingwithlovelace.hashnode.dev/part-1-first-look-at-the-first-program.

For $n = 1$ Operations 1–7 result in $v10 = 0$ after a single loop. The program then jumps to Operations 24 and 25, and ends. For $n = 2$, $v10$ is greater than zero after Operations 1–7. So the program engages Operations 8–12, resulting in $v10 = 0$ with Operation 12. It then jumps to Operations 24 and 25, then ends. For $n = 3$, the initial value of $v10$ is even larger. Operations 13–23 are thus included in the flow, followed by Operations 24 and 25, then the end. When $n > 3$, Operations 13–23 are looped, with the number of iterations of the loop increasing with n, and the program then ends after Operations 24 and 25. For example, Operations 13–23 loop twice for $n = 4$, three times for $n = 5$, and so on.[10] The process inspires a vision of ever-increasing Bernoulli numbers perpetually marching through the cogs and gears of a machine that rhythmically orchestrates their generation.

This series of calculations is a much simplified description of Lovelace's program. For me, the program demonstrates a practical application of Lovelace's media concept. Instead of in-betweenness, typified by Wordsworth and Babbage, we see Lovelace experimenting with polynomials in a technical operation that required remembering a large number of values from previous iterations of the process. This memory is impossible without the mechanisms of computer storage, which is itself inconceivable without values signified by numbers. The Bernoulli program leveraged Lovelace's distinctions among materiality, numerical abstraction, and rhythmic operation, while demonstrating their technical arrangement by the operation of the Analytical Engine.

Rhythm and Media Theory

I see my analysis above as engaging in what Matthew Kirschenbaum calls a "grammatology of the hard drive": a "technical and theoretical examination of the drive's basic operations—. . . which takes an instrument or device rather than a text as its locus" (19). While the tracing of a hard drive's operations is crucial for grammatological analysis, it is also important to recognize how abstraction enables usability for computing devices. Grammatology puts into relief Lovelace's distinctions among abstraction (numbers), the processes undertook by the object (rhythmic operation), and the material object (the device) while also showing how mediation orchestrates all three.

When Lovelace speculated that the Analytical Engine can operate on things besides number, she articulated a media concept that acknowledges these technical differences. Guillory identifies Walter Benjamin's article "The Work of Art in the Age of Mechanical Reproduction" (1935) as "the first great theorization of technical media" due to its analysis of the aura in art and the mass production

of film (347). Yet Lovelace's background in music and mathematics also gave her a means for theorizing the media concept with the Analytical Engine, which orchestrates distinct technical processes like a conductor orchestrates instruments. While Lovelace's scattered and brief reflections on the media concept hardly match the sophistication of Benjamin's essay, nevertheless they both demonstrate that media history might look more like what Siegfried Zielinski calls a variantology rather than a narrative of periodization or progression. Variantology favors attention to experiment, radical heterogeneity, and nonnormative media practices as an antidote to what Zielinski identifies as *psychopathia medialis*: the capitalist drive toward homogeneity and technological progressivism.[11] We can see the dialectic between variantology and homogeneity operating between Lovelace and Babbage. Lovelace appears as a media artist, leveraging material processes and mathematical concepts to speculate about the aesthetic possibilities of musical simulation. Babbage and other industrial capitalists saw these same concepts leading to the ever-granular modularization of bodies for surplus value. The historical evolution that Guillory sees as leading to the technical media concept might be better described as various forces struggling within this variantology: on the one hand, those who exploit media (and persons, resources, and material objects like the Analytical Engine) for money; and on the other hand, and as Zielinski says in his call to action at the end of *Deep Time of the Media* (2006), those who maintain "the right of access to those places which offer hospitality to [media] experimenters and experiments" (279).

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NOTES

1. Hollings, Martin, and Rice mention a governess of Ada Byron (Lovelace) whose notebook includes instructions for "lessons in the morning in . . . music . . . no more than a quarter of an hour long—after dinner . . . music . . . all performed with alacrity and docility" (qtd. in 9).

2. Toole mentions Lovelace's excitement "seeing the opera *Norma*, by Bellini, performed with Guilia Grisi playing the part of the high priestess (Ada later uses this term, high priestess, to describe her relationship to Babbage's plans for the Analytical Engine)" (91).

3. See Burkett; Menke; and Fyfe for more on nineteenth-century mediation.

4. The citation references Young's introduction to *Sonic Time Machines*, which nevertheless provides the most succinct definition of Ernst's concept in that source. Time-criticality defines media processes whose temporality is "processual and discrete, rather than static and continuous; . . . operational rather than narrative" (Young 13). As Young explains on the same page, Ernst's work in *Sonic Time Machines* is an elaboration and expansion of time-criticality through his use of the concept "sonicity," seen as an entanglement "between the time and frequency domains," though the former term appears rarely and is never explicitly defined in rest of the book. For more from Ernst on "time-critical media," see "From Media History to *Zeitkritik*."

5. See Whitson.

6. See "The Computer History Museum."

7. See Padua for more on the operations of the machine.

8. While Bernoulli discovered the underlying sequence of constants that would be named Bernoulli numbers, Faulhaber provided a formula Bernoulli used for calculating the sums of powers up to the 17th power. It can be expressed as $\sum_{k=1}^{n} k^p = 1^p + 2^p + \cdots + n^p = \frac{1}{p+1}\sum_{j=0}^{p}\binom{p+1}{j}B_j n^{p+1-j}$.

9. According to Laitinen, a decrement operator "subtracts one from the value of the variable" (151).

10. Alt notes that "2(13…23) means that the loop executes twice while ∑(+1) $n - 2$ is just a fancy way of saying it runs n-2 times."

11. See Zielinski for more on variantology and the need to recover the "great diversity" of media from *psychopathia medialis* and its standardized history (5).

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